M3C2 Simple Formula Guide

1 IQR and unusual values

Outlier means an unusual value

Most taxi rides cost 10 to 17. One ride costs 100. The value 100 is far from the others. It could be a mistake or a much longer ride.

Remember: Check the value before changing it.

Find the quarter marks

Put the numbers in order. This is the example from the lesson:

Values: 10, 11, 12, 13, 14, 15, 16, 17, 100

Mark	Simple meaning	Here
Q1	Lower quarter mark — 25% position	12
Median or Q2	Middle mark — 50% position	14
Q3	Upper quarter mark — 75% position	16

With the pandas method used in class, these are the 3rd, 5th and 7th values. A percentage here describes a position in the ordered list.

IQR means the width of the middle half

IQR stands for Interquartile Range. Subtract the lower quarter mark from the upper quarter mark.

$$IQR = Q_3 - Q_1 = 16 - 12 = 4$$

Use the IQR to find two check limits:

$$Lower\ limit = Q_1 - 1.5 \times IQR = 12 - 1.5 \times 4 = 6$$

$$Upper\ limit = Q_3 + 1.5 \times IQR = 16 + 1.5 \times 4 = 22$$

Flag values below 6 or above 22. Since 100 is above 22, we check it. Values exactly at a limit are not flagged.

Why 1.5? 1.5 is the usual multiplier in this rule. A larger multiplier gives wider limits and fewer flags.

Remember: A flag means “check this value”. It does not mean “delete this row”.

Lesson result: In the real Telco training data, there are no IQR flags in the three numeric columns.

Teacher note: quarter position = 1 + (count − 1) × fraction. Here: Q1 → 1 + 8 × 0.25 = 3; Q3 → 1 + 8 × 0.75 = 7. This matches the notebook method.

Average spread and z score

2 Find the centre and describe the spread

Average and median

Example: 10, 20, 90

$$Average = \frac{total}{number\ of\ values} = \frac{10 + 20 + 90}{3} = 40$$

The median is the middle value after sorting: here it is 20. With an even number of values, average the two middle values.

The large value 90 pulls the average up. The median stays at the middle of the list.

Standard deviation means spread around the average

Easy example: 10, 10, 20, 20. Their average is 15.

Step	What we do	Result
1	Subtract 15 from each value	−5, −5, 5, 5
2	Multiply each gap by itself	25, 25, 25, 25
3	Average the squares	100 ÷ 4 = 25
4	Take the square root	√25 = 5

Answer: Standard deviation = 5. It is a measure of spread, in the same units as the original values.

The average of the squared gaps is called variance. The square root of variance is the standard deviation.

Z score means how far from the average

$$z\ score = \frac{value - average}{standard\ deviation}$$

Use average 15 and standard deviation 5:

$$For\ 20: \frac{20 - 15}{5} = 1 \qquad For\ 10: \frac{10 - 15}{5} = -1$$

20 is one standard deviation above the average. 10 is one below it.

The lesson checks z-scores above 3 or below −3. The short form is |z| > 3: the bars mean “ignore the sign”.

In the notebook: In the IQR example, 100 has a z-score of about 2.82. IQR flags it; this z-score rule does not. The high value also increased the average and the spread.

Symbols you may see: μ = average; σ = standard deviation; x = a value. We divide by the number of values here, as the lesson does.

Two ways to change scale

3 Keep the same information and change the numerical scale

MinMax puts the training range between 0 and 1

Training values: 10, 20, 30. The minimum is 10 and the maximum is 30.

$$New\ \ value\ =\ \frac{value\ -\ training\ \ minimum}{training\ \ maximum\ -\ training\ \ minimum}$$

Example: For the value 20:

$$\frac{20\ -\ 10}{30\ -\ 10}\ =\ \frac{10}{20}\ =\ 0.5$$

Original value	After MinMax
10	0
20	0.5
30	1

Use the SAME minimum and maximum for a new value, 50:

$$\frac{50\ -\ 10}{30\ -\ 10}\ =\ 2$$

Remember: New values outside the training range can go above 1 or below 0.

StandardScaler uses the average and the spread

It uses the z-score calculation to change scale. The training average becomes 0, and one standard deviation becomes one unit.

$$New\ \ value\ =\ \frac{value\ -\ training\ \ average}{training\ \ standard\ \ deviation}$$

Easy example: Use 10, 10, 20, 20 from page 2: average = 15, standard deviation = 5.

Original value	Calculation	New value
10	(10 − 15) ÷ 5	−1
20	(20 − 15) ÷ 5	1

−1 means “one spread unit below the training average”. The original payment can still be positive.

Notebook check: 10, 20, 30 have average 20 and standard deviation about 8.165. Use the same formula.

Names to know: MinMax is often called normalization; StandardScaler does standardization. Both change scale. They do not remove outliers or create a bell-shaped distribution.

Distance neighbours and categories

4 How the model reads the inputs

Distance means how far apart two records are

Imagine two customer records. The gap in the first number column is 3. The gap in the second is 4.

$$Distance = \sqrt{(first\ \ gap)^2 + (second\ \ gap)^2}$$

$$Distance = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

For more number columns, add more squared gaps. Very large gaps can dominate this calculation. Scaling can change which records are closest.

KNN predicts from the nearest training examples

KNN means k nearest neighbours. In this lesson, k = 5: we use five neighbours.

Five neighbours say: Yes, Yes, No, Yes, No

Prediction: Yes. Three of the five neighbours say Yes.

Category encoding turns names into numbers

Integer coding gives each name a number. One-hot creates a separate yes/no column for each category: 1 means yes, 0 means no.

Colour	Integer code	Is red	Is green	Is blue
red	0	1	0	0
green	1	0	1	0
blue	2	0	0	1

Remember: The integer codes are labels. Code 2 does not mean that blue is twice as much as green.

With the lesson settings, an unseen category gets integer code −1, or zeros in all one-hot columns for that field.

Two graphs used in the lesson

Boxplot: The box goes from Q1 to Q3. The line inside is the median. The whiskers reach the last values inside the IQR limits; points beyond them are shown separately.

Histogram: Each bar counts how many values fall in one range.

Notebook words: tenure = months as a customer; MonthlyCharges = monthly payment; TotalCharges = total payment. customerID is an identifier, not a model input.

Check the result fairly

5 Use the same customers and the same learned rules

Accuracy means the share of correct predictions

$$Accuracy = \frac{correct\ \ predictions}{all\ \ predictions}$$

Example: The model gets 7 out of 10 answers right.

$$Accuracy = \frac{7}{10} = 0.7 = 70\,\%$$

In the lesson, compare the predicted Churn answer with the known Churn answer. Churn means the customer left or stayed.

Train validation and final test have different jobs

Data part	Job
Train	Train the model and learn the preparation rules
Validation	Compare the different methods
Final test	Make the final check later; keep it closed today

fit: learn a rule from train. For MinMax, learn the minimum and maximum.

transform: apply the saved rule to values.

fit_transform: learn the rule and apply it.

Example: Train contains 10, 20, 30. MinMax learns 10 and 30. If validation contains 50, keep those limits: the result is 2.

Learning the limits from validation would let the check data change the preparation. This is data leakage. Learn the rule from train, then apply it to validation.

Imputation means filling a missing value

Known training values are 10, 20, 30. Their median is 20. If we choose median filling, we replace a missing value with 20.

Remember: A blank means “unknown”. It is not automatically zero.

Fair comparison: Change one thing at a time: scaling in the first experiment; category coding in the second. Keep the customers, original fields and KNN settings fixed.

Lesson reference: M3C2_Starter.ipynb and Lesson Flow V2.

IQR reference | Quartile method | MinMax | StandardScaler